

Overview

This event provides recognition for FBLA members who understand security needs for technology.

Competencies

The topics listed below are prioritized, listing first the most important content areas of the tests.

- Firewalls
- Intrusion detection
- Network and physical security
- Cryptography
- Public key
- Authentication
- Computer attacks (e.g. virus, spam, spyware, Trojans, hijackers, worms)
- E-mail security
- Disaster recovery
- Forensic security

Procedures/Tips

- Review the Competitive Events Tips in the front of the book.

Web Site Resources

- <http://www.us-cert.gov/cas/tips/>
- <http://www.howstuffworks.com/firewall.htm>
- <http://www.garykessler.net/library/crypto.html>

Cyber Security Sample Questions

1. A(n) ____ needs to have sufficient processor speed and memory to handle the network's present traffic and increased traffic as the network grows.
 - a. bastion host
 - b. screening router
 - c. NAT server
 - d. three-pronged firewall
2. A(n) ____ is a client computer that is connected to the Internet and hosts firewall software.
 - a. screening router
 - b. dual-homed host
 - c. multi-firewall DMZ
 - d. DMZ screened subnet
3. A(n) ____ router filters traffic to individual computers within the internal network.
 - a. screening
 - b. PIX
 - c. blocking
 - d. reverse
4. A(n) ____ is a network that sits outside the internal network but is connected to the firewall and provides publicly available servers.
 - a. stub
 - b. area
 - c. perimeter
 - d. DMZ
5. The ____ server in the DMZ needs only list a limited number of public IP addresses.
 - a. Proxy
 - b. NAT
 - c. DNS
 - d. Firewall
6. A(n) ____ server is a server that creates a secure tunnel connection.
 - a. VPN
 - b. tunnel
 - c. authentication
 - d. RADIUS
7. A(n) ____ firewall is a device that monitors information going out of a network rather than trying to block what is coming in.
 - a. perimeter
 - b. reverse
 - c. NAT
 - d. access
8. Internet Connection Firewall (ICF) is a feature of the _____ operating system.
 - a. NetWare 6.x
 - b. Windows XP Professional
 - c. Mac OS X
 - d. Red Hat Linux 9.x

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9. Which of the following is **incorrect** about firewalls?
- a. restrict access to specific network resources
 - b. can be a hardware or software solution
 - c. create a protective layer between the network and the outside world
 - d. contain built-in filters
10. A _____ is a device that keeps certain kinds of network traffic out of a private network.
- a. Web server
 - b. firewall
 - c. cookie
 - d. compiler
11. What is the state when an IDS sensor or agent incorrectly identifies an attack as benign traffic called?
- a. checksum
 - b. false negative
 - c. anomaly
 - d. false positive
12. The IDS signature that is based on the packet's TCP or UDP port is called:
- a. NIDS
 - b. sensor signature
 - c. IP session logging
 - d. port signature
13. Your company has decided to put intrusion detection software on all servers and workstations. This is called:
- a. host-based intrusion detection
 - b. Lan-based intrusion detection
 - c. full-service detection
 - d. setting traps
14. A computer that is placed on a network to attract attackers is called a:
- a. lure or a trick
 - b. decoy or a honeypot
 - c. dummy or a firewall
 - d. trap or a snare
15. A process that runs in the background and performs a specified operation at predefined times is called a(n):
- a. operating system
 - b. daemon
 - c. firewall
 - d. SSH
16. Controlling access to critical network applications, data, and services is called:
- a. access control
 - b. perimeter security
 - c. security assessment
 - d. encryption

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17. The fire marshal in your city has just inspected the wiring that goes through the ceiling on the second floor of your building and says it is dangerous because the network wiring can cause toxic vapors in a fire. What is the wiring most likely to contain?
 - a. Teflon coating
 - b. PVC coating
 - c. rubber insulation
 - d. a hybrid metal jacket
 18. Any useful information about the target is called:
 - a. access
 - b. user information
 - c. threat
 - d. knowledge
 19. You have decided to use NTFS encryption to enhance security on your network of six servers. Five of the six servers have compressed drives and a new administrator says that it would not be a good idea to implement an encryption policy at this time. Why is or isn't the administrator correct?
 - a. The Administrator is correct. Encryption and compression cannot be used at the same time.
 - b. The administrator is correct. Encryption would add too much overhead to the servers.
 - c. The administrator is not correct. Encryption will help secure your network.
 - d. The administrator is not correct. Encryption will decrease the performance of the servers, but it will not be noticeable to the users.
 20. What is the published document called that explains to all users how the CA is structured?
 - a. certificate revocation list
 - b. certificate policy
 - c. certificate practice statement
 - d. certificate server
 21. A PKI (Public Key Infrastructure) document that serves as the vehicle on which to base common interoperability standards and common assurance criteria as an industry-wide basis is a certificate
 - a. process.
 - b. practice.
 - c. policy.
 - d. procedure.
 22. A(n) _____ threat is one that changes its apparent shape over time, representing a new threat not detectable by techniques that are looking for a preconfigured signature.
 - a. cataleptic
 - b. polymorphic
 - c. metamorphic
 - d. translucent
 23. An example of a good password is:
 - a. C73&gh2
 - b. admin
 - c. help
 - d. password
 24. When the target receives a SYN packet, it replies with a _____ packet.
 - a. TCP SYN ACK
 - b. SYN
 - c. TCP RST
 - d. TCP ACK

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25. When a virus infects the boot sector of a hard disk:
- the only other place it may infect is memory
 - it is common that disks placed in the floppy drive may become infected, too
 - the system generally boots faster than normal
 - print jobs run out of spool space on the hard disk
26. A _____ is not a virus, but an e-mail falsely warning of a virus.
- buffer attack
 - worm
 - virus hoax
 - Trojan horse
27. A mail user agent is:
- a server that determines how to send an e-mail message along the fastest route
 - a program used to compare an e-mail message and to read an e-mail message
 - software that transmits an e-mail message
 - an e-mail account
28. Because secure e-mail uses secret encryption keys that only the owners know and have access to, the recipient of the e-mail knows for a fact that it was sent by the person it purports to be from. This feature is called:
- authentication
 - nonrepudiation
 - integrity
 - confidentiality
29. _____ protects files and databases in case of an unexpected system crash or power failure by backing out of a data entry that is not fully completed before the system goes down, keeping the file from being corrupted.
- A standby UPS
 - Net Ware File Synchronization
 - Parallel Processing
 - The Transaction Tracking System
30. A(n) _____ is a device that provides power to electronic devices for a limited time period when the power goes out.
- UPS
 - RAID
 - USB
 - NIC

Cyber Security Answer Key

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|-------|-------|-------|
| 1) A | 11) B | 21) C |
| 2) B | 12) D | 22) B |
| 3) A | 13) A | 23) A |
| 4) D | 14) B | 24) D |
| 5) C | 15) B | 25) B |
| 6) B | 16) B | 26) C |
| 7) B | 17) B | 27) B |
| 8) B | 18) D | 28) A |
| 9) B | 19) A | 29) D |
| 10) B | 20) C | 30) A |

Database Design & Application Answer Key

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|-------|-------|-------|
| 1) D | 11) B | 21) C |
| 2) A | 12) A | 22) D |
| 3) C | 13) D | 23) C |
| 4) B | 14) D | 24) B |
| 5) D | 15) B | 25) A |
| 6) B | 16) C | 26) A |
| 7) D | 17) B | 27) D |
| 8) D | 18) B | 28) B |
| 9) D | 19) B | 29) B |
| 10) A | 20) A | 30) B |

Desktop Publishing Answer Key

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|-------|-------|-------|
| 1) D | 11) A | 21) C |
| 2) A | 12) A | 22) C |
| 3) C | 13) C | 23) C |
| 4) A | 14) C | 24) B |
| 5) A | 15) B | 25) B |
| 6) A | 16) B | 26) A |
| 7) B | 17) B | 27) B |
| 8) C | 18) D | 28) A |
| 9) B | 19) B | 29) A |
| 10) A | 20) C | 30) B |

Economics Answer Key

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|-------|-------|-------|
| 1) C | 11) A | 21) B |
| 2) D | 12) C | 22) D |
| 3) A | 13) D | 23) B |
| 4) C | 14) D | 24) B |
| 5) D | 15) C | 25) A |
| 6) D | 16) B | 26) B |
| 7) C | 17) D | 27) B |
| 8) B | 18) C | 28) B |
| 9) D | 19) A | 29) B |
| 10) A | 20) B | 30) C |

Entrepreneurship Answer Key

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|-------|-------|-------|
| 1) A | 11) A | 21) B |
| 2) B | 12) C | 22) B |
| 3) B | 13) B | 23) B |
| 4) B | 14) A | 24) A |
| 5) D | 15) D | 25) D |
| 6) A | 16) C | 26) C |
| 7) C | 17) B | 27) A |
| 8) B | 18) A | 28) A |
| 9) A | 19) B | 29) D |
| 10) B | 20) C | 30) B |